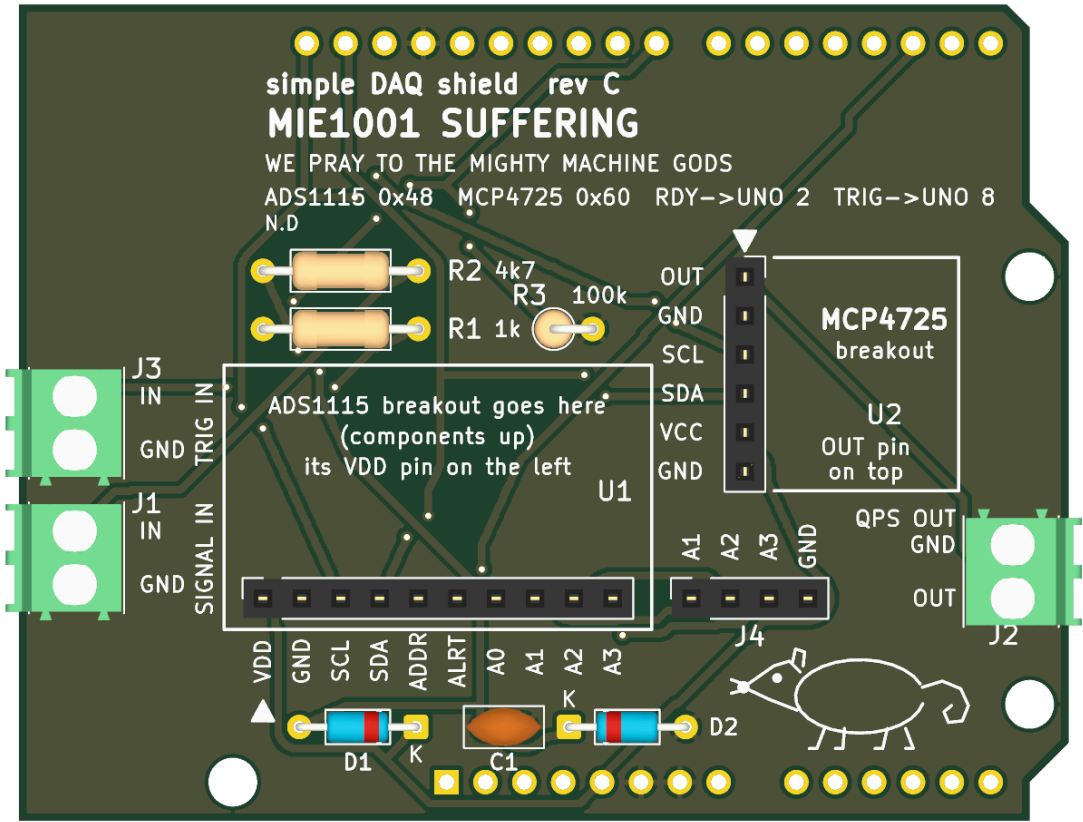


MIE1001 — simple DAQ shield (rev C)

The beginner version of the DAQ board: an Arduino UNO R3 shield that the two breakout boards from class plug into. Through-hole parts only, fat tracks, everything hand-solderable. JLCPCB only makes the bare board; you solder it yourselves.



SIGNAL IN (J1) -> R2 4k7 -> [D1/D2 clamp, C1 filter] -> ADS1115 breakout A0 -> I2C -> UNO -> USB -> laptop
UNO -> I2C -> SparkFun MCP4725 breakout OUT -> QPS OUT (J2), 0-5 V
TRIG IN (J3) -> R1 1k -> UNO D8 (R3 100k to GND) ADS1115 ALRT -> UNO D2 spare A1-A3 on J4

Check	Result
ERC	0 errors, 0 warnings
Independent critical review	Two reviews against the Adafruit, SparkFun, Arduino, Phoenix and TI source files and datasheets. Rev A: pinouts and UNO geometry correct, four fixes -> rev B. Rev C: copper clean, no respin needed; one wrong BOM part number and several wrong claims in this file, all corrected below

Still unverified	The height clearance between the underside of this shield and the UNO's ATmega chip. Nobody could find a dimensioned drawing: check it with the real boards
DRC incl. schematic parity	0 errors, 0 unconnected, 0 parity issues (4 <code>lib_footprint_mismatch</code> warnings: A1 and J1–J3 are deliberately modified copies)
Orientation	Same real-UNO geometry as the full board (rev E): flipped library footprint, keep-outs over the USB, DC jack and both ICSP headers
Rules	0.5 mm tracks, 0.3 mm gaps, 0.8/0.4 mm vias: far above JLCPCB's minimums

What changed in rev C

Three parts on the signal input, in the free band under the ADS1115 socket:

- **C1 (100 nF) to ground.** With R2 it is a single-pole low-pass at **339 Hz**, and it supplies the charge the ADC's switched-capacitor input takes at each sample. It takes the edge off fast noise and limits the bandwidth, so leave it out if you need to measure something faster than ~300 Hz. **It does not remove mains hum:** at 50 Hz it attenuates by only 1.1 %. See the firmware notes for what actually works. **C1 must be X7R or C0G, not Y5V** (a Y5V part loses most of its value when warm, which moves the corner frequency anywhere up to ~2 kHz).
- **D1 and D2 (1N4148) clamps.** They divert most of an over-voltage to the rails. Be clear about what does the work: **R2 is what keeps the chip legal** (1.3–2.4 mA at ± 12 V against its 10 mA limit), because the 1N4148 only starts conducting around 0.5–0.7 V, by which point the input is already past the ADC's absolute maximum of $V_{DD} + 0.3$ V and the chip's own internal protection diode is conducting alongside D1. The diodes are there for the energy of a static zap or a badly wrong connection, not to hold the pin inside spec.
- **What the diodes cost:** up to 25 nA of leakage each, which through R2 is up to 117 μ V of offset (15 counts at GAIN_SIXTEEN, and several times worse when hot). Fine at low gain; calibrate the offset if you use the high gain ranges.
- **One side effect:** with the Uno unpowered, a signal above 5 V feeds about 1.3 mA back into its 5 V rail through D1. Don't leave a live preamp connected to an unpowered board.
- **No amplifier.** The ADS1115 has a programmable gain amplifier built in, so a small signal is handled in software with `ads.setGain(...)` instead of another chip. Note that at the highest gains the chip's differential input resistance falls to about 710 k, where R2 costs ~0.66 % of gain (0.02 % at the ± 6.144 V range): calibrate if you use GAIN_SIXTEEN. Note TI suggests keeping a filter resistor under 1 k; 4.7 k is deliberate here, chosen for protection, and C1 at the pin supplies the sampling charge that the series resistance otherwise would.

Watch the stripe when soldering the diodes. The silkscreen prints **K** at the cathode end of each one, and that is the end with the black band on the part. D1's K points right, towards +5 V. D2's K points left, towards the signal. Backwards, they short the signal to a rail.

What changed in rev B

- **R2 (4k7) in series with SIGNAL IN.** The ADS1115 input may not go above $V_{DD} + 0.3$ V (TI datasheet, absolute maximum ratings). If the preamp swings to its ± 12 V rails, R2 keeps the current into the chip's input clamp to 1.3–2.4 mA (its limit is 10 mA). Gain error: 0.02 % on the ± 6.144 V range, 0.66 % on the ± 0.256 V range.

- **R3 (100k) pull-down on D8.** A TRIG IN with nothing connected now reads LOW instead of picking up noise and giving false triggers.
- **Pin-1 triangles and "which way round" text on both sockets.** Both breakouts fit in their sockets rotated 180°. For the MCP4725 that puts 5 V on its GND pin. Now the MCP4725 outline is the real 15.2 × 15.2 mm board.
- **Firmware and BOM fixes.** `dac.begin(0x60)` (the library default is 0x62), and the BOM no longer orders the new two-row ADS1115 board that doesn't fit.

Before you order: three things to check by hand

1. **Which ADS1115 board do you have?** Use the one from the class kit; do not order a new one. This shield fits the **original** Adafruit board with **one row of 10 pins** (VDD GND SCL SDA ADDR ALRT A0 A1 A2 A3). Adafruit now sells a newer STEMMMA QT version (two rows of 6, "VIN" instead of "VDD") under the same product number 1085, and **it does not fit**. Your team's self-test sketch says "VDD", so you probably have the original, but count the pins.
2. **Print `fab/print_1to1_top.pdf` at 100 %** and lay the UNO and both breakouts on it. The header holes must line up with the UNO's pins and with each breakout's pins.
3. **Height over the UNO's ATmega chip.** D1, D2, C1, the ADS1115 socket and J4 all sit above it. Lying the parts flat and trimming the leads flush should clear it, but this has not been verified against a drawing: check it with the real UNO before soldering everything down.

Parts (full list in `BOM.csv`)

Ref	Part	Where
U1	Adafruit ADS1115 breakout (original, 1×10), from the class kit	plugs into a 1×10 female header (Sullins PPTC101LFBN-RC, Digi-Key S7008-ND). Don't order 1528-1085-ND: that is the new two-row board
U2	SparkFun MCP4725 breakout BOB-12918	plugs into a 1×6 female header (PPTC061LFBN-RC, S7004-ND). Retired at SparkFun, still at Digi-Key 1568-1047-ND
J1, J2, J3	Phoenix 1984617 screw terminals, 3.5 mm	Digi-Key 277-1721-ND
J4	1×4 female header for the spare inputs	S7002-ND
R1	1k ¼ W resistor on the trigger	1.0KQBK-ND
R2	4k7 ¼ W resistor in series with SIGNAL IN	4.7KQBK-ND
R3	100k ¼ W pull-down on D8, mounted standing up	100KQBK-ND
C1	100 nF 50 V X7R disc, 2.5 mm lead pitch (not Y5V: BC1160-ND is the wrong one)	BC1084CT-ND
D1, D2	1N4148 signal diodes (DO-35)	1N4148FS-ND
—	UNO stacking headers	Adafruit 85 (legs just clear the USB jack)

No other parts. Both breakouts already carry the I²C pull-ups: the ADS1115 board has 10k on SDA and on SCL (hard-wired), the MCP4725 board has 4.7k on each through jumper SJ1 (closed by a trace from the factory). In parallel that is 3.2k per line, about 2.9k once the Arduino's own internal pull-ups are counted, which draws 1.6 mA at the low level against the 3 mA an I²C device must sink. The minimum sensible value here is about 1.5k, so this is fine and the shield adds no pull-ups of its own. If you ever chain more I²C boards onto this one, each brings its own pull-ups and the total drops: cut SJ1 on the MCP4725 board to take its 4.7k pair off the bus. The ADS1115 breakout also pulls ALRT (D2) up with 10k.

Ordering and building

1. JLCPCB: upload `fab/mie1001_daq_simple_gerbbers.zip`, 2 layers, 1.6 mm, HASL is fine. No assembly.
2. Solder, lowest parts first: D1, D2 (lying flat, **stripe towards the K on the board**), R1 and R2 (lying flat), C1, R3 (standing up, body on the circle), then the female headers (keep them straight: plug the breakout in while soldering), the terminal blocks, and finally the stacking headers.
3. **Trim every lead under the board flush. This is required, not optional.** The ADS1115 socket and J4 sit right above the UNO's ATmega chip, and D1, D2 and C1 are in the same band. An untrimmed socket lead sticks out about 1.6 mm and a terminal pin about 2.9 mm. Check the clearance over the ATmega with the real boards before you solder everything down.
4. Plug the breakouts in **component side up**, matching the pin names on the breakout to the names on the shield: VDD on the left for the ADS1115, and OUT at the top (at the ▼) for the MCP4725. Rotated 180° the MCP4725 gets 5 V on its GND pin.

Firmware notes

- ADS1115 at **0x48**, MCP4725 at **0x60**. Libraries: Adafruit ADS1X15 and Adafruit MCP4725.
- `dac.begin(0x60)`: the Adafruit_MCP4725 library defaults to 0x62, so a plain `dac.begin()` finds nothing. `ads.begin()` (0x48) is correct as is.
- Read the signal on A0. `ads.setGain(GAIN_TWOTHIRDS)` gives ± 6.144 V full scale. For a small signal turn the gain up instead of adding an amplifier: `GAIN_SIXTEEN` gives ± 0.256 V full scale and ~ 7.8 μ V steps.
- **The input is filtered at 339 Hz** (R2 with C1), so the board cannot follow anything faster. Leave C1 out if you need the bandwidth.
- **For 50 Hz mains hum, use the ADC, not the filter.** Set a low data rate (`ads.setDataRate(RATE_ADS1115_8SPS)`), whose internal averaging window rejects mains, or average a batch of readings in the sketch. C1 barely touches 50 Hz.
- **J4's spare inputs (A1–A3) have no resistor, clamp or filter.** They go straight to the chip. Keep them under 0–VDD, and tie them to GND if you don't use them.
- **The useful input range is 0 V to VDD** (the USB 5 V, often only 4.8 V). This board expects a voltage from Team 3's preamp, not the raw detector current. R2 protects the chip from a preamp that overshoots, but readings above VDD are not valid.
- The MCP4725 powers up at mid-scale (2.5 V on QPS OUT). Write 0 to its EEPROM once.
- Trigger in: `pinMode(8, INPUT)`, active HIGH. R3 holds D8 low when nothing is connected. Use 5 V logic if you can: the ATmega needs 3.0 V for a HIGH, so 3.3 V logic has only 0.3 V to spare. R1 limits the current if the trigger goes a little above 5 V.

- ALRT goes to **UNO pin 2**: `pinMode(2, INPUT)` is enough, the breakout has the pull-up. (Careful reading the board: the silkscreen "D2" next to C1 is the *diode* D2, not the UNO's pin 2.)

Compared with the full board (`../mie1001_daq_board`, rev E)

This one has no electrometer amplifier and no guard ring: it digitises a voltage that someone else has already made. Since rev C it does have basic input protection (R2 with D1/D2) and a 339 Hz filter (R2 with C1), but nothing that can measure a current. That is exactly what the class breakouts are for, and it is much easier to build and debug. For picoamp detector currents, use the full board.

Files

`generate_schematic.py` and `generate_pcb.py` generate everything (the PCB script reuses the UNO geometry and the router of the full board); `./build.sh` rebuilds and re-runs all checks. The possum and the prayer to the machine gods are in `generate_pcb.py`, `possum()`.